

RAJARAM MOHAN BALAJI

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PROFILE

Data Engineering & Cloud Computing master's student who builds production-grade systems, not just notebooks. In the past year shipped a real estate price prediction platform live on Render (Docker, GitHub Actions CI/CD, FastAPI), and built a multi agent AI analyst routing queries across 630+ vector embeddings. Strong in Python, SQL, ETL pipeline design, and containerized deployment. Seeking a data engineering or ML platform role to keep building systems that run in production, not just in a demo.

KEY TECHNICAL SKILLS

- Languages:** Python, Java, SQL
- Data Engineering:** ETL Pipelines, Data Modelling, Data Warehousing, Data Quality
- Databases:** MySQL, NoSQL | Advanced SQL: window functions, CTEs, subqueries
- Core CS:** OOP, Data Structures & Algorithms, Database Management
- Visualisation:** Power BI, Plotly, Streamlit, KPI dashboards, EDA, data storytelling
- Soft Skills:** Problem Solving • Team Collaboration • Adaptability • Fast Learner • Analytical Thinking

PROJECTS

- AivancityX EdTech Admin Dashboard (Ongoing)** | Python, Streamlit, Plotly
- Developing a full featured Streamlit admin dashboard for Aivancity as a real college client project
 - Built live KPI widgets and interactive Plotly charts to visualise student and course performance metrics
- OpenHousing : Real Estate Price Prediction Platform** | Python, FastAPI, scikit-learn, Docker, GitHub Actions
github.com/Mohan790/openhousing | Live: openhousing-8q7c.onrender.com/app/
- Built an end to end ML pipeline: automated ETL and a Random Forest model ($R^2 = 0.88$, $MAE \approx \$2,000$) deployed live on Render with Docker and a GitHub Actions CI/CD pipeline
 - Developed a FastAPI REST API with a real-time web interface; shipped per IP rate limiting after a community reported production security review
- Enterprise AI Data Analyst** | Python, LangGraph, Qdrant, FastAPI, Docker
github.com/Mohan790/enterprise-ai-analyst
- Built a multi modal ReAct agent (LangGraph + Llama 3.1) routing business questions between SQL and semantic vector search over 10 financial reports
 - Engineered a Vector ETL pipeline: semantically chunked documents, embedded with HuggingFace, and stored 632+ vectors in a Dockerized Qdrant database
- FreshCheck AI** | Python, PyTorch, Streamlit, MySQL
github.com/Mohan790/food-quality-inspector
- Fine tuned MobileNetV2 on 13,000+ real world food images, achieving 99.01% validation accuracy with a custom PyTorch training loop and Grad CAM explainability
 - Deployed a live Streamlit app with real-time Fresh/Rotten classification, confidence charts, and heatmap overlays

PROFESSIONAL EXPERIENCE

- Web Development Intern — Scorpleo Technologies** | Sep 2024 – Dec 2024 | Coimbatore, India
- Integrated dynamic SQL driven data retrieval into production web interfaces; contributed to code reviews and Git based collaborative workflows
 - Shipped front-end and back-end features using HTML5, CSS, Python, and SQL directly on live client projects
- Trainee Associate, Healthcare IT — Baruch Business Solutions** | Aug 2023 – Jul 2024 | Coimbatore, India
- Wrote and optimised SQL queries to extract, clean, and validate patient datasets, reducing manual reporting effort for the team
 - Managed daily processing of patient records across healthcare information systems, maintaining data accuracy and regulatory compliance

EDUCATION

- MSc Data Engineering & Cloud Computing — Aivancity, Paris 2025 – Present
- BSc Information Technology — Dr. N.G.P. Arts & Science College 2020 – 2023

CERTIFICATIONS

- Master Data Management for Beginners — TCS iON • AI Capabilities and Limitations — Anthropic • Big Data Foundations — IBM
- MySQL & PHP Development — Spoken Tutorial

LANGUAGES

- Tamil: Native • English: Professional (C1) • French: Beginner (B1)